

Hochschild Homology:

classical, topological, & motivic

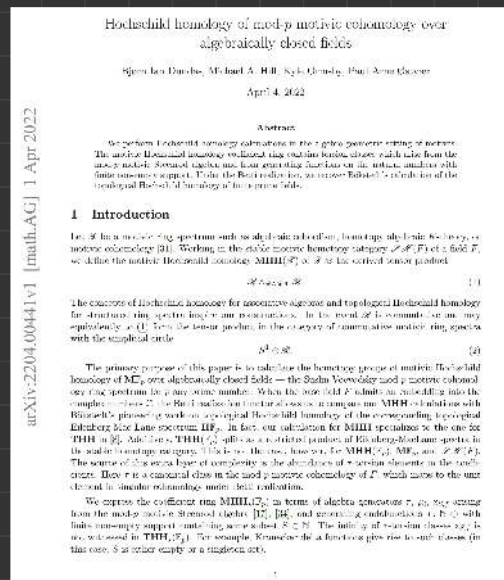


USC Algebra Seminar, 15.IV.22
Kyle Ormsby, Reed College



Outline

- ① Hochschild homology
- ② Topological & motivic variants
- ③ Computations over $\mathrm{Spec} \mathbb{C}$
- ④ Two truths and a lie?



All work joint with Bjørn Dundas, Mike Hill, & Paul Arne Østvær



① Hochschild homology

Setup: k a field, A a commutative k -algebra

Hochschild complex ($\otimes = \otimes_k$):

$$\begin{aligned} \textcircled{\star} \quad \cdots \rightarrow A^{\otimes n+1} &\rightarrow A^{\otimes n} \rightarrow \cdots \rightarrow A^{\otimes 2} \rightarrow A \\ a_0 \otimes a_1 \otimes \cdots \otimes a_n &\mapsto \sum_{i=0}^{n-1} (-1)^i a_0 \otimes \cdots \otimes a_{i-1} \otimes a_i a_{i+1} \otimes a_{i+2} \otimes \cdots \otimes a_n \\ &\quad + (-1)^n a_n a_0 \otimes a_1 \otimes \cdots \otimes a_{n-1}. \end{aligned}$$

E.g. $a \otimes b \otimes c \mapsto ab \otimes c - a \otimes bc + ca \otimes b.$

Defn $HH_n(A/k) = H_n(\textcircled{\star}) = \text{Tor}_n^{A \otimes A}(A, A)$

derived
version of
 $A \otimes_{A \otimes A} A$

Hochschild - Kostant - Rosenberg Theorem

Suppose $\text{char } k = 0$ and A is smooth of finite type over k .

Then $\underbrace{HH_*(A/k)} \simeq \underbrace{\Omega_{A/k}^*}$.

de Rham complex: $\Omega_{A/k}$ corepresents k -linear derivations on A ($\text{Hom}_k(\Omega_{A/k}, M) \cong \text{Der}_k(A, M)$),
 $\Omega_{A/k}^n = \Lambda_k^n \Omega_{A/k}$ with "exterior derivative" differential

→ Does $HH_*(A/k)$ carry a natural differential compatible with de Rham?

Yes, coming from an S' -action.

Going in Circles

Recall $\underbrace{HH(A/k)}_{\text{Hochschild complex, previously } \textcircled{*}} = A \otimes_{A \otimes A}^{\mathbb{L}} A$. By Dold-Kan, this is the

geometric realization of some simplicial commutative k -algebra.

Fact $HH(A/k) = \left| \text{pushout} \left(\begin{array}{c} A \amalg A \rightarrow A \\ \downarrow \\ A \end{array} \right) \right|$
each $A \in [\Delta^{\text{op}}, \text{CAlg}_k]$ constant

$= |S' \otimes A|$, using ① $[\Delta^{\text{op}}, \text{CAlg}_k]$ is tensored over $[\Delta^{\text{op}}, \text{Set}]$,
② $S' = \text{pushout} \left(\begin{array}{c} \delta^0 \rightarrow * \\ \downarrow \\ \mathbb{I} \end{array} \right)$.

Going in Circles (ct'd)

So $S' \otimes A$ carries an action of S' and pushed through
Dold-Kan, we get an action of $C^*(S', k)$ on $HH(A/k)$
 $\Rightarrow H_*(S'; k) \cong k[\varepsilon]/(\varepsilon^2)$ acts on $HH_*(A/k)$.
 $|\varepsilon| = 1$

The "Connes differential" is $HH_n(A/k) \longrightarrow HH_{n+1}(A/k)$.
 $\eta \longmapsto \varepsilon \cdot \eta$

Under the hypotheses of HKR, this corresponds to the de Rham differential.

② Topological Hochschild Homology

/ sphere spectrum!

Instead of working over k (or $\mathbb{Z}$), let's work over $\mathbb{S}$.

"Just" as one has

$$\begin{array}{ccccc} \mathbb{Z}\text{-mod} & \xrightleftharpoons[-\text{res}]{-\otimes_{\mathbb{Z}} \mathbb{R}} & \mathbb{R}\text{-mod} & \xrightleftharpoons[-\text{res}]{-\otimes_{\mathbb{R}} \mathbb{C}} & \mathbb{C}\text{-mod} \end{array}$$

we also have

$$\begin{array}{ccc} -\otimes_{\mathbb{S}} \mathbb{Z} = -\wedge H\mathbb{Z} & \text{derived category of Abelian groups} & \\ \text{Sp} & \xrightleftharpoons[H]{\quad} & D(\mathbb{Z}) \end{array}$$

↳ symmetric monoidal ∞ -category of spectra,
 $\otimes = \wedge$; constructed from pointed spaces
 by inverting suspension $\Sigma = S^1 \wedge -$.

Topological Hochschild Homology (ct'd)

Every spectrum X has Abelian homotopy groups $\pi_n X$, $n \in \mathbb{Z}$.

For $C \in \mathcal{D}(\mathbb{Z})$, $\pi_n \mathbb{H}C = H_n(C)$.

Given a commutative ring object $A \in \mathcal{D}(\mathbb{Z})$ (usual ring, dga, ...)

Bökstedt defined

$$T\mathbb{H}(A) := \mathbb{H}A \wedge_{\mathbb{H}A \wedge \mathbb{H}A} \mathbb{H}A = S' \otimes \mathbb{H}A.$$

(In fact, may form $T\mathbb{H}(E_\infty\text{-ring})$ in this fashion.)

Then $T\mathbb{H}_n(A) := \pi_n T\mathbb{H}(A)$.

By Serre finiteness, if $A \cong \mathbb{Q}$, then $T\mathbb{H}_*(A) \cong \mathbb{H}_*(A/\mathbb{Z})$.

Applications

① Improved de Rham theory in characteristic p .

Thm (Hesselholt) k a perfect field of char $p > 0$, R a smooth k -algebra, then $THH_n(R) \cong \bigoplus_{i \geq 0} \Omega_{R/k}^{n-2i}$ as R -modules.

Work of Bhatt-Morrow-Scholze et al in p -adic Hodge theory.

② Trace methods: $\exists$ Dennis/Bökstedt trace

$$\underbrace{K(R)} \longrightarrow THH(R)$$

algebraic K-theory spectrum of R

which factors through $TC(R)$, the topological cyclic homology of R . Dundas-McCarthy: cyclotomic trace is an "infinitesimal p -adic equivalence".

Computations

Since $THH(A) = HA \wedge_{HA \wedge HA} HA$ and $\pi_* HA = A$ (in $\deg 0$),

we have a Tor-spectral sequence

$$E_{h,t}^2 = \text{Tor}_{h,t}^{\pi_* HA \wedge HA} (A, A) \Rightarrow THH_{h+t}(A)$$

$$\text{and } d^r : E_{h,t}^r \rightarrow E_{h-r, t+r-1}^r.$$

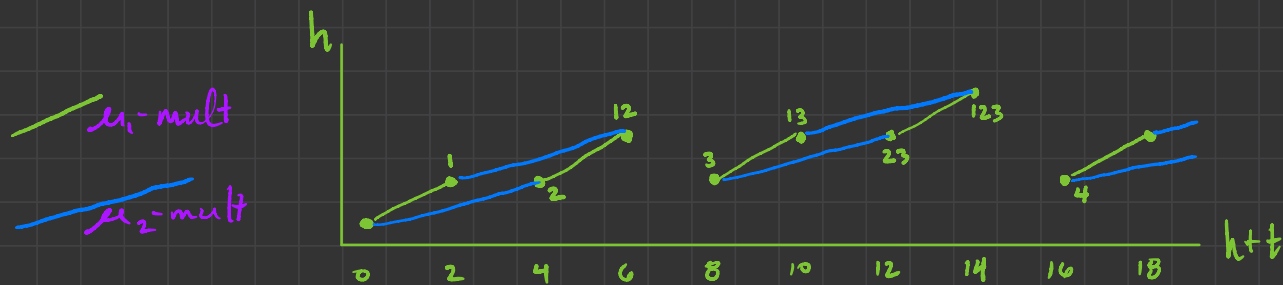
For $A = \mathbb{F}_2$, $\pi_* H\mathbb{F}_2 \wedge H\mathbb{F}_2 = \mathcal{A}_* = \mathbb{F}_2[\xi_1, \xi_2, \xi_3, \dots]$, $|\xi_i| = 2^i - 1$
is the mod-2 dual Steenrod algebra.

Thus $E^2 \cong \bigwedge_{\mathbb{F}_2} (\mu_1, \mu_2, \mu_3, \dots)$ with $|\mu_i| = (1, 2^i - 1)$.

Pictorially ...

Computations (ct'd)

$$E^2 \cong \bigwedge_{\mathbb{F}_2} (\mu_1, \mu_2, \mu_3, \dots) \quad \text{with } |\mu_i| = (1, 2^i - 1)$$



Note ① One class in each even degree.

② Differentials go 1 left & down, so $E^2 = E^\infty$.

Fact By a power operations argument, $\mu_i^2 = \mu_{i+1} \in THH_*(\mathbb{F}_2)$.

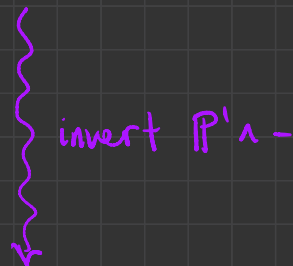
Thm (Bökstedt) $THH_*(\mathbb{F}_p) = \mathbb{F}_p[\mu]$, $|\mu| = 2$ for all primes p .

<new> Motivic Hochschild Homology

Now move from spectra to motivic spectra and replace HA with MA , the motivic Eilenberg-Mac Lane spectrum.

Morel-Voevodsky:

$[S_{\mathcal{M}/k}, [\Delta^{op}, Set]] + \text{Nisnevich \& } \mathbb{A}^1\text{-localize} = \text{motivic spaces } \mathcal{S}pc_k$



motivic spectra $\mathcal{S}p_k$

Motivic Hochschild Homology (ct'd)

$$\begin{array}{ccc}
 A' \setminus 0 & \longrightarrow & A' \simeq * \\
 \downarrow \quad \uparrow & & \downarrow \\
 * \simeq A' & \longrightarrow & P'
 \end{array}
 \quad \Rightarrow \quad P' \simeq S' \wedge (A' \setminus 0) .$$

simplicial circle
geometric circle

Upshot Bigraded spheres $S^{m,n} := (S')^{m-n} \wedge (A' \setminus 0)^{\wedge n}$

total degree
weight

Thus we have bigraded homotopy groups

$$\pi_{m,n} X := [S^{m,n}, X]$$

for $X \in \mathcal{S}p_k$.

Note Need homotopy sheaves to detect weak equivs.

Motivic Hochschild Homology (ct'd)

Important (co)homology theories are representable in $\mathcal{S}p_k$:

- MA = motivic cohomology, with coefficients in A
- KGL = (homotopy) algebraic K-theory
- KQ = Hermitian K-theory
- MGL = algebraic cobordism

We define the motivic Hochschild homology of a commutative ring A to be the motivic spectrum

$$MHH(A) := MA \wedge_{MA \wedge MA} MA = S' \otimes MA$$

with coefficients $MHH_{*,*}(A) = \bigoplus_{m,n \in \mathbb{Z}} \pi_{m,n} MHH(A)$.

③

Computations over $\mathbb{C}$

Fix k algebraically closed, $A = \mathbb{F}_p$, $p \neq \text{char}(k)$.

Theorem 1.1. Over an algebraically closed field of exponential characteristic $e(F) \neq p$, there is an algebra isomorphism

$$\mathbf{MHH}_*(\mathbb{F}_p) \cong \mathbb{F}_p[\tau, \mu_i, x_{S,f}]_{i \in \mathbb{N}, (S \subset \text{supp } f, f: \mathbb{N} \rightarrow \mathbb{N})} / \mathcal{I} \quad (3)$$

with the ideal of relations

$$\mathcal{I} = \left(\begin{array}{l} \mu_i^p - \tau^{p-1} \mu_{i+1}, \\ \tau^{p-1} x_{S,f}, \\ x_{S,f} \cdot x_{T,g} - \sum_{u \in \text{supp}(f+g) - S \cup T} \epsilon_u \cdot x_{S \cup T \cup \{u\}, f+g} \end{array} \right).$$

Here the support of f is a finite non-empty subset of the natural numbers and $S \subset \text{supp } f \subset \mathbb{N}$ does not contain the minimal element of $\text{supp } f$. The coefficient $\epsilon_u \in \mathbb{F}_p$ is given explicitly in Definition 2.12. The algebra generators have bidegrees given by $|\tau| = (0, -1)$, $|\mu_i| = (2p^i, p^i - 1)$, and

$$|x_{S,f}| = (|S| + 1)(-1, p - 1) + p \sum_{j \in \text{supp } f} f(j)(2p^j, p^j - 1).$$

③

Computations over $\mathbb{C}$ (ct'd)

Fix k algebraically closed, $A = \mathbb{F}_p$, $p \neq \text{char}(k)$.

Theorem 1.1. Over an algebraically closed field of exponential characteristic $\text{e}(F) \neq p$, there is an algebra isomorphism

$$\text{MHH}_*(\mathbb{F}_p) \cong \mathbb{F}_p[t, \mu_i, x_{S,f}]_{i \in \mathbb{N}, (S \subset \text{supp } f, f: \mathbb{N} \rightarrow \mathbb{Z})} / \mathcal{I} \quad (3)$$

with the ideal of relations

$$\mathcal{I} = \left(\begin{array}{l} \mu_i^p - p^{p-1} \mu_{i+1}, \\ p^{p-1} x_{S,f}, \\ \sum_{u \in \text{supp}(f+g) - S \cup T} \epsilon_{u, S, T, \{u\}, f+g} x_{S, f} \cdot x_{T, g} \end{array} \right).$$

Here the support of f is a finite non-empty subset of the natural numbers and $S \subset \text{supp } f \subset \mathbb{N}$ does not contain the minimal element of $\text{supp } f$. The coefficient ϵ_u is given explicitly in Definition 2.12. The algebra generators have bidegrees given by $|v| = [0, -1]$, $|\mu_i| = [2p^i, p^i - 1]$, and

$$|x_{S,f}| = (|S| + 1)(-1, p - 1) + p \sum_{j \in \text{supp } f} f(j)(2p^j, p^j - 1).$$

Definition 2.12. For functions $f, g: \mathbb{N} \rightarrow \mathbb{Z}$ with finite support and non-empty finite sets $S, T \subseteq \mathbb{N}$ define $K_{S,T,f,g} \in \mathbb{F}_p$ by

$$K_{S,T,f,g} = \left(\prod_{s \in S} \binom{fs - 1 + gs}{fs - 1} \right) \left(\prod_{t \in T} \binom{ft + gt - 1}{ft} \right) \left(\prod_{c \notin S \cup T} \binom{fc + gc}{fc} \right)$$

if $(S, f), (T, g) \in J$ and $S \cap T = \emptyset$, and set $K_{S,T,f,g} = 0$ otherwise. Moreover, we define

$$\epsilon_{u, S, T, f, g} = K_{S \cup \{u\}, T \cup \{t_{f+g}\}, f+g} + K_{S \cup \{t_{f+g}\}, T \cup \{u\}, f+g}.$$

Yikes! Goals ① Shape of the computation when $p=2$.
 ② Consequences in motivic homotopy.

Computation Strategy

Input $\pi_{**} \mathbf{MF}_2 = \mathbb{F}_2[\tau], |\tau| = (0, -1)$

$$\mathcal{A}_{**} = \pi_{**} \mathbf{MF}_2 \wedge \mathbf{MF}_2 \cong \mathbb{F}_2[\tau, \xi_1, \xi_2, \dots, \tau_0, \tau_1, \dots] / (\tau_i^2 - \tau \sum_{i \geq 0} \xi_{i+1} \mid i \geq 0)$$

$$|\xi_i| = (2^{i+1} - 2, 2^i - 1), \quad |\tau_i| = (2^{i+1} - 1, 2^i - 1)$$

Step 1 Calculate étale motivic Hochschild homology

$$\mathrm{MH}_{**}(\mathbb{F}_2)[\tau^{-1}] \cong \mathbb{F}_2[\tau^{\pm 1}, \mu_0, \mu_1, \dots] / (\mu_i^2 - \tau \mu_{i+1} \mid i \geq 0)$$

$$\cong \mathrm{THH}_{**}(\mathbb{F}_2)[\tau^{\pm 1}].$$

Step 2 Calculate mod- τ MHH

$$\mathrm{MH}_{**}(\mathbb{F}_2) / \tau \cong \bigwedge_{\mathbb{F}_2} (\bar{\mu}_0, \bar{\mu}_1, \dots) \otimes \bigwedge_{\mathbb{F}_2} (\bar{\lambda}_1, \bar{\lambda}_2, \dots).$$

divided powers algebra

Computation Strategy

Step 3 τ -torsion in $MHH_{**}(\mathbb{F}_2)$ injects into $MHH_{**}(\mathbb{F}_2)/\tau$ with image that of the τ -Bockstein.

Step 4 Give a presentation of τ -torsion in $MHH_{**}(\mathbb{F}_2)$ in terms of generators $x_{S,f}$ where $f: \mathbb{N} \rightarrow \mathbb{N}$ has finite support and $S \subseteq \text{supp } f$.

Step 5 Combine étale and τ -torsion computations via a pullback square.

Step 1 Étale MHH

$$\begin{aligned} \pi_{**} MF_2 \wedge MF_2[\tau^{\pm 1}] &\cong \mathbb{F}_2[\tau^{\pm 1}, \xi_1, \xi_2, \dots, \tau_0, \tau_1, \dots] / (\tau_i^2 - \tau_{i+1} \mid i \geq 0) \\ &\cong \mathbb{F}_2[\tau^{\pm 1}, \tau_0, \tau_1, \dots], \quad |\tau_i| = (2^{i+1}-1, 2^i-1) \end{aligned}$$

so the Tor-spectral sequence takes the form

$$\begin{aligned} E^2 &= \operatorname{Tor}_{*,*,*}^{\mathbb{F}_2[\tau^{\pm 1}, \tau_0, \tau_1, \dots]}(\mathbb{F}_2[\tau^{\pm 1}], \mathbb{F}_2[\tau^{\pm 1}]) \\ &\cong \bigwedge_{\mathbb{F}_2}(\mu_0, \mu_1, \dots)[\tau^{\pm 1}] \Rightarrow MHH_{**}(\mathbb{F}_2)[\tau^{\pm 1}] \end{aligned}$$

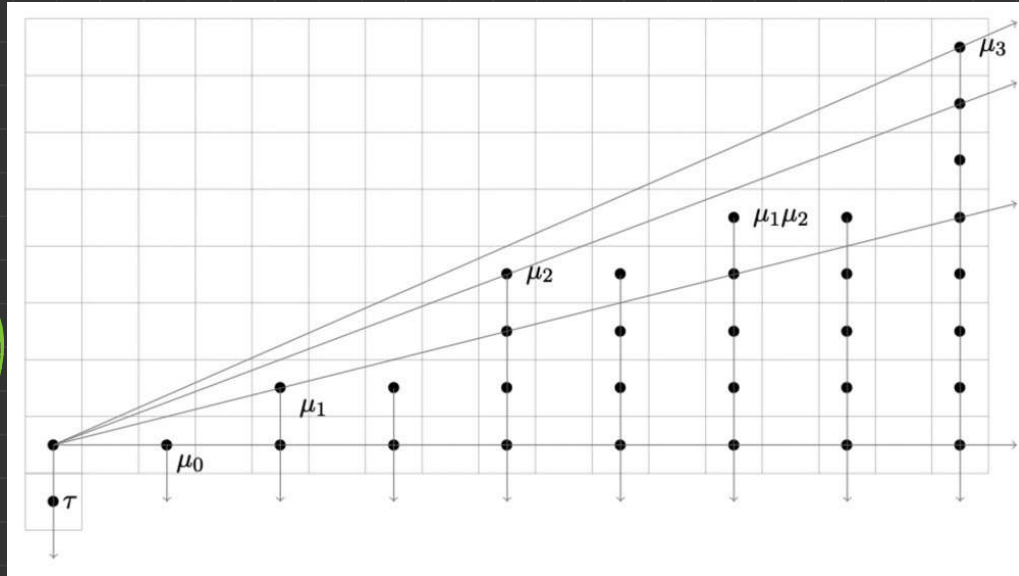
with $|\mu_i| = (1, 2^{i+1}-1, 2^i-1)$.

- Degree considerations $\Rightarrow E^2 = E^\infty$.
- Power operations $\Rightarrow \mu_i^2 = \tau \mu_{i+1}$.

Step 1 Étale MHH (ct'd)

$$\begin{aligned} \underline{\text{Upshot}} \quad \text{MHH}_{**}(\mathbb{F}_2)[\tau^{-1}] &\cong \mathbb{F}_2[\tau^{\pm 1}, \mu_0, \mu_1, \dots] / (\mu_i^2 - \tau \mu_{i+1} \mid i \geq 0) \\ &\cong \mathbb{F}_2[\tau^{\pm 1}, \mu_0] \end{aligned}$$

im(MHH_{**}($\mathbb{F}_2$)
 $\rightarrow$ (") $[\tau^{-1}]$)



Step 2 Mod- τ MHH

$A_{**}/\tau \cong \mathbb{F}_2[\bar{\xi}_1, \bar{\xi}_2, \dots] \otimes \Lambda_{\mathbb{F}_2}(\tau_0, \tau_1, \dots)$ so the Tor-spectral sequence takes the form

$$E^2 = \text{Tor}_{*,*,*}^{A_{**}/\tau}(\mathbb{F}_2, \mathbb{F}_2) \cong \Lambda_{\mathbb{F}_2}(\bar{\lambda}_1, \bar{\lambda}_2, \dots) \otimes \underbrace{\Gamma_{\mathbb{F}_2}(\bar{\mu}_0, \bar{\mu}_1, \dots)}_{\text{divided powers algebra}} \Rightarrow \text{MHH}_{**}(\mathbb{F}_2)/\tau.$$

Advanced degree yoga $\Rightarrow E^2 = E^\infty$ with no hidden ext's.

Upshot

$$\text{MHH}_{**}(\mathbb{F}_2)/\tau \cong \Lambda_{\mathbb{F}_2}(\bar{\lambda}_1, \bar{\lambda}_2, \dots) \otimes \Gamma_{\mathbb{F}_2}(\bar{\mu}_0, \bar{\mu}_1, \dots).$$

(See arXiv: 2204.0041 for Steps 3-5.)

④ Consequences

Since $THH(\mathbb{F}_p)$ is an $H\mathbb{F}_p$ -module, we get the splitting

$$THH(\mathbb{F}_p) \simeq \bigvee_{i \geq 0} \Sigma^{2i} H\mathbb{F}_p.$$

The τ -torsion in $MHH_{**}(\mathbb{F}_p)$ implies that this fails wildly in $Sp_{\mathbb{C}}$:

$MHH(\mathbb{F}_p)$ is not a free $M\mathbb{F}_p$ -module.

In Sp , the following are true:

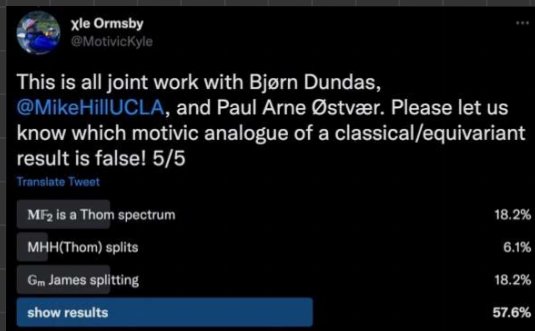
- ① $H\mathbb{F}_2$ is a Thom spectrum of an E_2 -map with target $\Omega^2 S^3$
- ② $THH(\text{Thom}_1) = \text{Thom}_2 = \text{Thom}_1 \wedge B(\text{base}_1)_+$
- ③ $B\Omega^2 S^3 \simeq V(\text{spheres})$ stably $\eta_{\text{top}} = 0$

Consequences (ct'd)

Potential motivic analogues:

- ① MF_2 is a Thom spectrum over $\Omega^{2,1}S^{3,1}$.
- ② $MHH(Thom_1) \simeq Thom_1 \wedge B(base_1)_+$.
- ③ $B\Omega^{2,1}S^{3,1}$ is a wedge of spheres stably,
i.e. $\Sigma^\infty \Omega^{1,1} \Sigma^{1,1} S^2$ satisfies " G_m -James splitting".

Thm (Dundas, Hill, Ø, Østvær) At most two of these are true!



Thank You!



Questions welcome.